

PROJECT SUMMARY

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Executive Project Summary: 180 Ton 3-Storey Steel Frame

DATE: October 26, 2023 PROJECT REF: S-23-451 PREPARED BY: STRUCTMIND AI

Project Description This project entails the detailing, fabrication, and erection of a 180-ton structural steel frame for a new three-story building. The structure's lateral stability is achieved through a system of steel moment frames, requiring specialized moment connections at critical beam-to-column joints. The design, materials, and workmanship will adhere strictly to the AISC 360-22 Specification for Structural Steel Buildings and the AISC 341-22 Seismic Provisions, as applicable. The scope includes all primary columns, beams, and girders required to complete the structural skeleton for subsequent trades.

Key Quantities

- Total Tonnage: 180 tons (A992 Grade 50 steel)
- Total Pieces (Est.): 2,000 – 2,300 members
- Primary Moment Connections (Est.): 160 – 200 locations (requiring CJP welds or high-strength bolting)
- Simple Shear Connections (Est.): 1,100 – 1,400 locations

Top 5 Risks 1. Connection Quality & Inspection: Moment connections demand a high level of execution and rigorous inspection (e.g., Ultrasonic Testing). Insufficient welder qualification or inspection bottlenecks can cause significant schedule delays and costly rework. 2. Erection Tolerances: The geometric control and plumbness of the moment frames are critical for fit-up and structural performance. Failure to maintain specified tolerances can cascade through the structure. 3. MEP Integration: Deep moment frame beams and connection components can create conflicts with mechanical, electrical, and plumbing systems. Late-stage coordination failures will result in expensive field modifications. 4. Skilled Labor Shortage: The availability of certified field welders and ironworkers experienced with complex moment frame erection is a primary constraint that could impact both schedule and budget. 5. Material Lead Times: Specific heavy wide-flange sections required for columns or long-span beams may have extended mill rolling schedules, delaying the start of fabrication.

Top 5 Opportunities 1. Connection Optimization: Value engineer the specified moment connections. Exploring proprietary bolted systems (e.g., SidePlate, DuraFuse) or alternative welded details could drastically reduce field labor and inspection time. 2. Prefabrication / Modularization: Maximize

shop-welded sub-assemblies, such as "column trees," to minimize complex and weather-dependent field welding, improving quality and accelerating the erection schedule. 3. Early Steel Procurement: Lock in material pricing by placing mill orders upon approval of advanced bill of materials, mitigating exposure to steel price volatility. 4. Advanced BIM Modeling: Utilize a highly detailed (LOD 400) structural model to conduct clash detection with MEP and architectural designs, resolving all potential conflicts before fabrication begins. 5. Phased Erection Planning: Develop a detailed erection sequence that stabilizes bays sequentially, allowing concrete and MEP trades to begin work on lower floors while steel erection continues above.

Recommended Schedule Horizon A total duration of 14-18 weeks is recommended from the release of approved-for-construction drawings to the completion of steel erection.

- Detailing & Submittal Approval: 3-4 Weeks
- Material Procurement & Fabrication: 6-8 Weeks
- Erection & Final Inspection: 5-6 Weeks